

Notes: Benmelech, Garmaise, and Moskowitz (QJE, 2005)

Do Liquidation Values Affect Financial Contracts? Evidence from Commercial Loan Contracts and Zoning Regulation

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1 Big picture

- **Question.** Do higher **asset liquidation values** lead creditors to offer **more debt, cheaper debt, longer debt, and simpler creditor structures**?
- **Setting.** The paper studies **commercial real estate loan contracts** in the United States and uses **zoning regulation** to proxy for how easily a property can be redeployed to another use if the lender forecloses.
- **Core challenge.** Liquidation value is central in incomplete-contracting theories, but it is very hard to observe *ex ante*. Realized recovery values are endogenous to default states and market conditions, so the paper needs a precontract measure of collateral quality.
- **Main empirical idea.** Within a local area and broad zoning category, properties with more flexible zoning can be used in more alternative ways. That makes them more **redeployable**, which the authors interpret as a higher **next-best-use value** and therefore a higher liquidation value.
- **Main contribution.** The paper connects incomplete-contracting theory directly to contract-level evidence by showing that more redeployable assets receive:
 - lower interest rates,
 - higher leverage,
 - longer maturity and duration,
 - fewer multiple-creditor structures.
- **Why it matters.** The paper is really about **collateral quality**, not just collateral quantity. It shows that the liquidation value of an asset changes both the **price** and the **structure** of financing.
- **Theoretical predictions tested.** The paper organizes the evidence around five predictions from Williamson, Harris–Raviv, Hart–Moore, Shleifer–Vishny, Bolton–Scharfstein, and Diamond:
 1. debt levels rise with liquidation value;
 2. promised yields fall with liquidation value, conditional on debt level;
 3. debt maturity rises with liquidation value;
 4. multiple creditors are more likely when liquidation value is low;
 5. current asset value rises with liquidation value.
- **Main empirical findings.**
 - More redeployable properties receive **larger** and **longer** loans at **lower interest rates**.
 - The probability of **multiple creditors** falls when redeployability is high.
 - The baseline results survive **census tract fixed effects**, **bank fixed effects**, and rich controls for current price and earnings.
 - The effects are **stronger where foreclosure is easier**, **stronger where zoning rules are stricter**, and **stronger where local land markets are tighter**.

- **Historic zoning** works in the opposite direction, lowering loan supply and raising borrowing costs.
- **Economic takeaway.** Lenders care not only about the cash flow currently generated by a property, but also about what the property is worth **in the lender's hands after liquidation**. Better redeployability increases effective collateral value and expands debt capacity.

2 Data and measurement (key objects)

2.1 Transaction and loan-level data

- **Source.** The main data come from **COMPS.com**, a commercial real estate transaction database.
- **Sample.** The sample contains **14,159 commercial property transactions** from **January 1, 1992, to March 30, 1999** across **11 states plus the District of Columbia**.
- **What is observed for each transaction.**
 - sale price,
 - exact zoning designation,
 - debt frequency (whether a loan is present),
 - leverage / loan-to-value,
 - interest rate,
 - maturity,
 - amortization,
 - whether the rate is floating,
 - whether the loan is Small-Business-Administration-backed,
 - number of creditors,
 - latitude and longitude.
- **Additional linked data.**
 - Census data,
 - Wharton Land Use Control Survey,
 - crime risk data from CAP Index, Inc.
- **Basic magnitudes from Table I.**
 - Mean sale price: about **\$2.39 million**.
 - Mean debt frequency: about **0.71**.
 - Mean leverage: about **0.71**.
 - Mean loan rate: about **8.28%**.
 - Median maturity (duration): about **15 years (6.8 years)**.
 - Mean property age: just under **29 years**.

- **Why this market is useful.** Commercial real estate loans are usually **secured** and often **nonrecourse**, so creditor recovery depends heavily on the collateral itself. That makes liquidation value especially important.

2.2 Zoning designations and redeployability

- **Zoning categories.** The sample contains **161 unique zoning designations**, grouped into broad categories such as:
 - organizations,
 - waterfront,
 - manufacturing,
 - residential,
 - business,
 - commercial,
 - commercial/manufacturing,
 - historic.
- **Two dimensions of zoning.**
 - A **general category** (e.g., residential or manufacturing).
 - A **within-category intensity** that determines how broad the permitted use is inside that category.
- **Key measurement idea.** Higher within-category numeric values correspond to **broader allowable uses**. A property with a more flexible zoning assignment can be redeployed more easily after foreclosure.
- **Main liquidation-value proxy.** The paper converts the within-category zoning intensity into a **scaled redeployability measure**. Higher values mean more flexibility and a higher likely value in the next best use.
- **Scaling.** The within-zoning value is scaled by the **maximum allowable within-category value in the local jurisdiction**. The baseline scaling geography is the **zip code**, though the results are robust to other local definitions.
- **Distribution.** Figure I shows that the redeployability measure has:
 - mean **0.51**,
 - median **0.50**,
 - standard deviation **0.24**,
 - range **0.08 to 1.00**.

2.3 Property value and earnings controls

- **Sale price.** The transaction sale price proxies for **value in current use**.

- **Cap rate.** The paper uses the **capitalization rate**:

$$\text{CapRate}_i = \frac{\text{net income on the property in the previous year}}{\text{sale price}_i}.$$

- **Why this matters.** The authors want the zoning variable to capture **secondary value / liquidation value**, not simply current profitability or current market value.

2.4 Local market definition and fixed effects

- **Main identifying geography.** The regressions rely heavily on **census tract fixed effects**.
- **Why census tracts are useful.** A census tract is much smaller than the geography at which zoning laws are set or local lending markets typically operate. So within a tract, financing conditions are relatively similar, while properties can still differ in zoning flexibility.
- **Sample breadth.** The data contain **2,090 census tracts**; many tracts have multiple observations, allowing within-tract comparisons.
- **Interpretation.** The identifying variation is therefore a comparison of **otherwise nearby and locally similar properties** that differ in how flexibly they are zoned.

3 Models and empirical specifications (all equations + interpretation)

3.1 Theory predictions from incomplete contracting

The paper is mainly empirical, but the empirical design is organized around a set of theoretical predictions. Let LV denote liquidation value. Then the theory implies:

$$\begin{aligned} \frac{\partial \text{Debt Level}}{\partial LV} &> 0, & \frac{\partial \text{Promised Yield}}{\partial LV} &< 0 \text{ (conditional on debt level),} \\ \frac{\partial \text{Maturity}}{\partial LV} &> 0, & \frac{\partial \text{Pr(Multiple Creditors)}}{\partial LV} &< 0, & \frac{\partial \text{Current Asset Value}}{\partial LV} &> 0. \end{aligned}$$

Interpretation. Better collateral in the lender’s liquidation state makes the loan safer. That raises the amount lenders are willing to supply, lowers the required interest rate, makes long-term debt more feasible, and reduces the need to use multiple creditors as a disciplining device.

3.2 Redeployability measure

For property p in general zoning category A and jurisdiction j , the paper’s redeployability proxy can be written as

$$\text{Redeployability}_p = \frac{n_p}{\max_{q \in P(A,j)} n_q},$$

where n_p is the property’s within-category zoning intensity and $P(A, j)$ is the set of properties in category A in jurisdiction j .

Interpretation.

- If Redeployability_p is high, the property can be put to many alternative uses within its broad zoning class.
- If Redeployability_p is low, the property is tightly constrained in potential uses.
- The paper interprets this difference as a difference in **ex ante liquidation value**.

3.3 Loan duration construction

The paper computes loan duration from the loan interest rate, maturity, and amortization schedule, under the assumptions that:

- coupons are paid annually,
- the loan may have amortization,
- there is one final balloon payment at maturity.

Interpretation. Duration is included because the theory is about more than just face maturity. A higher liquidation value can also support **longer effective exposure** for creditors.

3.4 Baseline regression specification

The main empirical specification can be summarized as

$$y_i = \beta \text{Redeployability}_i + \gamma_1 \log(\text{price}_i) + \gamma_2 \text{CapRate}_i + \Gamma' X_i + \alpha_{\text{tract}(i)} + \alpha_{\text{zoning}(i)} + \alpha_{\text{type}(i)} + \alpha_{\text{year}(i)} + \varepsilon_i,$$

where y_i is one of:

- loan interest rate,
- debt frequency,
- leverage,
- debt maturity,
- loan duration,
- multiple-creditor indicator.

X_i includes additional controls such as local banking concentration, log property age, crime risk, and for interest-rate regressions also leverage, floating-rate status, SBA backing, maturity, and amortization.

Interpretation.

- $\beta < 0$ for the interest rate and multiple-creditor regressions means more redeployable assets get **cheaper** and **simpler** debt.
- $\beta > 0$ for leverage, maturity, and duration means more redeployable assets get **more debt** and **longer debt**.
- Controlling for price and cap rate is meant to isolate the collateral/liquidation-value channel from current-use value.

3.5 Bank fixed effects specification

To show the results are not driven by certain banks specializing in certain types of properties, the paper also estimates

$$y_i = \beta \text{Redeployability}_i + \gamma_1 \log(\text{price}_i) + \gamma_2 \text{CapRate}_i + \Gamma' X_i + \alpha_{\text{bank}(i)} + \alpha_{\text{tract}(i)} + \alpha_{\text{zoning}(i)} + \alpha_{\text{type}(i)} + \alpha_{\text{year}(i)} + \varepsilon_i.$$

Interpretation. This compares **two properties financed by the same bank** in similar neighborhoods. If the coefficient on redeployability survives, the result is less likely to be driven by bank selection.

3.6 Validation interactions

The paper tests whether redeployability matters more when liquidation value should matter more. A generic interaction specification is

$$y_i = \beta \text{Redeployability}_i + \eta (\text{Redeployability}_i \times Z_j) + \Gamma' X_i + \text{fixed effects} + \varepsilon_i,$$

where Z_j is one of:

- foreclosure cost,
- zoning strictness,
- growth-management effectiveness,
- local demand-to-supply tightness.

Interpretation.

- If redeployability really proxies for liquidation value, its effect should be **weaker when foreclosure is costly**.
- It should be **stronger when zoning is actually enforced**.
- It should also be **stronger when alternative uses are economically valuable**, i.e., when local land markets are tight.

3.7 Historic-zoning and price regressions

The paper also estimates a historic-zoning specification:

$$y_i = \beta_1 \text{Historic}_i + \beta_2 \text{Redeployability}_i + \beta_3 (\text{Redeployability}_i \times \text{Historic}_i) + \Gamma' X_i + \text{fixed effects} + \varepsilon_i.$$

Interpretation. Historic zoning is especially restrictive and hard to change, so it should reduce redeployability and make current zoning more binding.

Finally, the paper tests whether market prices rise with liquidation value using

$$\log(\text{price}_i) = \beta \text{Redeployability}_i + \gamma \text{current earnings}_i + \text{fixed effects} + \varepsilon_i.$$

Interpretation. This is a direct test of the prediction that current asset value rises with liquidation value. The authors are more cautious about causal interpretation here than in the contract regressions.

4 Identification and instruments (what makes the estimates credible)

4.1 What makes the identification hard

- Liquidation value is usually not directly observed at the time the contract is written.
- Realized recovery values are contaminated by default states, market conditions, and bargaining.
- Local zoning and local lending conditions may both reflect omitted neighborhood characteristics.

Core problem. A simple cross-sectional correlation between zoning and loan terms would not by itself show that liquidation value drives financing.

4.2 Main identification strategy

- The paper compares properties **within the same census tract**.
- It also compares properties **within the same broad zoning category**.
- It controls for **sale price** and **cap rate**.
- It adds **property type**, **year**, and eventually **bank fixed effects**.

Identification intuition. After absorbing local credit-market conditions and current-use value, the remaining within-tract, within-category variation in zoning flexibility is interpreted as variation in **collateral redeployability**.

4.3 Why zoning is a plausible liquidation-value proxy

- Zoning changes the set of legally permissible uses of the property.
- More permitted uses mean more potential buyers or more fallback uses after seizure.

- Rezoning and variances are typically costly and uncertain, so zoning can be binding in practice.

Economic intuition. This is exactly the Williamson / Shleifer–Vishny idea: collateral is more valuable when it can be sold or repurposed more easily.

4.4 Evidence against alternative explanations

- **Local financing environment.** The authors show redeployability within a tract is not significantly related to several lending-bank characteristics.
- **Bank specialization.** Adding **bank fixed effects** leaves the main results largely intact.
- **Current profitability.** Controlling for $\log(\text{price})$ and cap rate is meant to separate current-use value from liquidation value.
- **Neighborhood “quality.”** The paper notes that if restrictive zoning simply captured higher-quality neighborhoods, the sign of the results would be hard to reconcile with the estimates.

4.5 Validation from cross-sectional heterogeneity

The strongest evidence that the zoning variable is really about liquidation value is that its effects behave exactly as the theory predicts:

- **Foreclosure costs.** Redeployability matters *less* where foreclosure is slow and costly.
- **Zoning strictness.** Redeployability matters *more* where zoning rules are harder to change and more strictly enforced.
- **Local market tightness.** Redeployability matters *more* when demand for land uses is high relative to zoned supply.
- **Historic zoning.** Extremely restrictive zoning shifts contract terms in the opposite direction.

Interpretation. These patterns are hard to explain with a purely current-cash-flow or neighborhood-quality story and strongly support the liquidation-value channel.

4.6 What is still less clean

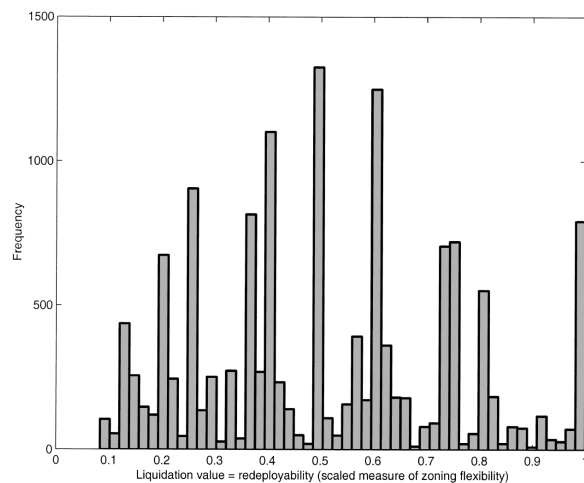
- The market-price regression is less clean because price and zoning may be jointly determined.
- The empirical strategy is strongest for the **loan-contract results**, which rely on very fine local comparisons and multiple validation tests.

5 Tables and figures

5.1 Figure I: distribution of redeployability

- **Read.** The redeployability measure is broadly dispersed, with many properties in the interior of the support rather than at only a few zoning bins.

- **What it shows.** The variable has economically meaningful variation: mean = 0.51, median = 0.50, standard deviation = 0.24, range 0.08 to 1.00.
- **Why it matters.** The paper is not just comparing a handful of restrictive and unrestricted parcels. There is continuous variation in the liquidation-value proxy.



Summary statistics of the liquidation value measure standard					
	Mean	Median	Standard deviation	Minimum	Maximum
Redevelopability	0.51	0.50	0.24	0.08	1

FIGURE I

Distribution of Redevelopability (Zoning Flexibility)

The distribution of a measure of real asset liquidation value determined by a proxy for the asset's redeployability measured by its zoning classification is plotted, below. The allowable use of the property within its broad zoning category and local zoning jurisdiction, scaled by the maximum allowable uses within an area and zoning category, is the measure of redeployability. Higher values indicate broader scopes of allowable uses within a general category and jurisdiction.

5.2 Table I: summary statistics

- **Read.** This table describes the sample by zoning category and by property type.
- **Main facts.**
 - Debt is used in roughly **71%** of transactions.
 - Average leverage is about **71%**.
 - The average sale price is about **\$2.39 million**.
 - Residentially zoned properties have especially long maturities.
 - Historic properties are relatively expensive and have comparatively tight financing.
- **Why it matters.** The sample looks like a market where debt capacity is important and where different zoning regimes plausibly matter for collateral value.

Table I, Panels A–B

TABLE I
SUMMARY STATISTICS OF ZONING DESIGNATIONS, COMMERCIAL REAL ESTATE
TRANSACTIONS, AND PROPERTY TYPES

PANEL A: MEAN CHARACTERISTICS OF PROPERTIES ACROSS GENERAL ZONING CATEGORY							
Zoning category	Number	Debt frequency	Leverage	Price	Maturity (duration)	Loan rate	Multiple creditors
ALL PROPERTIES	14,159	0.71	0.71	2,386,767	15 (6.8)	8.28	0.12
Organizations (O)	311	0.63	0.72	3,495,907	10 (7.9)	8.25	0.10
Waterfront (W)	6	0.67	0.85	4,887,500	15 (8.6)	7.00	0.25
Manufacturing (M)	3,188	0.68	0.72	1,807,378	10 (6.8)	8.73	0.13
Residential (R)	7,917	0.81	0.74	1,404,530	25 (10.0)	7.84	0.13
Business (B)	1,827	0.67	0.72	3,478,963	7 (6.4)	8.65	0.07
Commercial (C)	4,878	0.68	0.67	3,138,222	10 (6.9)	8.64	0.12
Comm/Manu. (CM)	252	0.74	0.74	1,003,192	10 (6.6)	8.74	0.19
Historic (H)	258	0.68	0.66	3,581,531	10 (7.9)	9.08	0.13

PANEL B: DISTRIBUTION OF ZONING CATEGORY ACROSS PROPERTY TYPE							
General zoning type (abbreviated), number of properties							
Property type	O	W	M	R	B	C	CM
Retail	94	2	227	247	837	1,898	87
Commercial	35	0	107	127	218	749	31
Industrial	20	0	1,953	44	78	230	68
Apartment	28	0	253	5,860	110	383	12
Mobile home park	1	0	1	19	0	2	0
Special	10	0	5	176	18	47	3
Residential land	38	0	37	1,160	14	57	1
Industrial land	5	0	362	16	3	16	4
Office	74	4	227	233	520	1,396	38
Hotel	6	0	16	35	29	100	8

Table I, Panel C

PANEL C: MEAN CHARACTERISTICS OF PROPERTIES ACROSS PROPERTY TYPE
(CONTINUED)

Property type	Number	Debt frequency	Leverage	Price	Maturity (duration)	Loan rate	Multiple creditors	Cap rate
Retail	3,949	0.74	0.72	1,610,357	10 (6.6)	8.80	0.10	10.33
Commercial	1,650	0.40	0.68	1,670,517	4 (4.9)	8.97	0.07	10.38
Industrial	3,784	0.70	0.73	1,589,490	10 (6.7)	8.72	0.12	9.97
Apartment	6,997	0.90	0.74	1,529,293	25 (10.0)	7.77	0.13	10.04
Mobile home park	41	0.76	0.71	5,087,748	10 (6.8)	8.46	0.19	9.19
Special	290	0.70	0.77	2,109,284	10 (6.2)	8.88	0.20	11.00
Residential land	1,713	0.41	0.75	1,004,216	7 (4.1)	8.91	0.11	N/A
Industrial land	568	0.37	0.74	921,757	8 (4.8)	9.06	0.08	N/A
Office	3,380	0.67	0.68	6,595,045	10 (6.6)	8.60	0.10	10.17
Hotel	270	0.63	0.69	10,574,474	14 (6.6)	8.82	0.22	12.21

Panel A reports the average loan frequency, loan-to-value (LTV) ratio, sale price, median loan maturity and duration (in parentheses) in years, loan rate (percent per year), frequency of multiple lenders (second/subordinated loans), and number of unique zoning code designations for all properties and for each general zoning category. Panel B reports the distribution of general zoning categories across ten property types. The number of properties under each of the eight broad zoning categories for each property type are reported. Panel C reports the average loan frequency, loan-to-value (LTV) ratio, sale price, loan maturity (in years), loan rate (percent per year), frequency of multiple lenders (second/subordinated loans), and capitalization rate (net income on the property in the previous year divided by

5.3 Table II: baseline loan-contract evidence

- **Read.** Table II is the core regression table. Panel A uses census tract fixed effects; Panel B adds bank fixed effects.
- **Main baseline coefficients from Panel A.**
 - Interest rate: $\beta = -0.6311$.
 - Debt frequency: $\beta = 0.0078$ (not significant).
 - Leverage: $\beta = 0.0447$.
 - Debt maturity: $\beta = 2.4821$.
 - Loan duration: $\beta = 0.4892$.
 - Multiple creditors: $\beta = -0.0926$.
- **Economic interpretation.**
 - More redeployable assets borrow **more cheaply**.
 - Conditional on borrowing, they borrow **more**.
 - They receive **longer maturity** and **longer duration** loans.
 - They are **less likely** to use multiple creditors.
 - The only weak margin is **whether there is any debt at all**; the action is mainly in contract terms conditional on financing.
- **Bank fixed effects.** Panel B shows the same-bank comparison delivers very similar estimates, which is a major credibility check.

ASSET REDEPLOYABILITY (MEASURED BY ZONING INTENSITY OF USE)
AND DEBT CONTRACTS

PANEL A: CENSUS TRACT FIXED EFFECTS						
Dependent variable =	Interest rate	Debt frequency	Leverage	Debt maturity	Loan duration	Multiple creditors
Redeployability	-0.6311 (-2.59)	0.0078 (0.13)	0.0447 (2.12)	2.4821 (1.94)	0.4892 (2.50)	-0.0926 (-2.36)
log(price)	-0.0850 (-3.85)	-0.0235 (-4.67)		-0.7173 (-5.94)	-0.0678 (-3.65)	0.0091 (2.61)
Cap rate	0.0081 (1.98)	0.0077 (8.01)	0.0042 (2.60)	0.2292 (10.11)	0.0393 (11.24)	-0.0027 (-4.16)
<i>Fixed effects?</i>						
Census tract	yes	yes	yes	yes	yes	yes
General zoning	yes	yes	yes	yes	yes	yes
Property type	yes	yes	yes	yes	yes	yes
Year	yes	yes	yes	yes	yes	yes
R^2	0.64	0.35	0.34	0.51	0.46	0.27
R^2 (no F.E.)	0.26	0.08	0.06	0.16	0.10	0.04
# Observations	3,536	9,365	7,733	7,733	1,971	7,733

PANEL B: CENSUS TRACT AND BANK FIXED EFFECTS						
Dependent variable =	Interest rate	Debt frequency	Leverage	Debt maturity	Loan duration	Multiple creditors
Redeployability	-0.8121 (-4.08)	0.0271 (0.59)	0.0477 (2.31)	2.0535 (1.21)	0.6679 (2.82)	-0.0964 (-2.04)
log(price)	-0.0963 (-3.86)	-0.0321 (-7.04)		-0.4951 (-2.81)	-0.0489 (-1.90)	0.0320 (4.41)
Cap rate	0.0280 (5.85)	0.0051 (5.99)	0.0024 (1.57)	0.1111 (3.60)	0.0327 (7.62)	-0.0002 (-0.15)
<i>Fixed effects?</i>						
Bank	yes	yes	yes	yes	yes	yes
Census tract	yes	yes	yes	yes	yes	yes
General zoning	yes	yes	yes	yes	yes	yes
Property type	yes	yes	yes	yes	yes	yes
Year	yes	yes	yes	yes	yes	yes
R^2	0.86	0.42	0.59	0.67	0.73	0.86

Panel A reports regression results of the loan interest rate, frequency of debt, total leverage, debt maturity, loan duration, and the frequency of multiple creditors on a measure of real asset redeployability, using the allowable use of the property given by its zoning classification. Additional regressors include the log of the sale price of the property (excluded from the loan-to-value regression), the capitalization rate of the property (the current earnings on the property divided by the sale price), the Herfindahl index of banking concentration within a fifteen-mile radius of the property, the log of property age, and the current crime risk level and recent growth rate in crime risk for the property's location (obtained from CAP Index, Inc.). The interest rate regressions also include the leverage ratio, an indicator for floating rates, an indicator for whether the loan is backed by the Small Business Administration, and the loan maturity and amortization as regressors. Regressions include fixed effects for general zoning category, property type, year, and census tract. Regressions are run under OLS with robust standard errors. Coefficient estimates and their associated t -statistics (in parentheses) are reported along with adjusted R^2 's, including and excluding the fixed effects, and the number of observations. Panel B adds bank fixed effects to the regressions.

5.4 Tables III and IV: cross-sectional validation and historic zoning

- **Table III read.** This table asks whether redeployability matters more in settings where liquidation value should matter more.
- **Table III punchline.**
 - Redeployability matters less when **foreclosure is costly**.
 - It matters more when **zoning is stricter**.
 - It matters more where **growth-management tools are effective**.
 - It matters more when **demand for land uses is high relative to supply**.
- **Interpretation.** These are exactly the comparative-statics one would expect if zoning flexibility is capturing **liquidation value** rather than an omitted current-cash-flow characteristic.
- **Table IV read.** Historic zoning is used as an alternative, very restrictive measure of low redeployability.
- **Table IV punchline.**
 - Historic designation raises interest rates by about **59 basis points**.
 - It lowers the probability of debt by about **10.8 percentage points**.

- It lowers leverage by about **4.9 percentage points**.
- It raises the probability of multiple creditors by about **11.9 percentage points**.
- **Interpretation.** Properties with especially inflexible, hard-to-change uses face meaningfully tighter credit.

Table III

CROSS-SECTIONAL EVIDENCE ON REDEPLOYABILITY AFFECTING DEBT CONTRACTS						
Dependent variable =	Interest rate	Debt frequency	Leverage	Debt maturity	Loan duration	Multiple creditors
PANEL A: INTERACTIONS WITH FORECLOSURE COSTS						
Redeployability	-1.4389 (-4.11)	-0.0083 (-0.10)	0.0362 (1.24)	4.8318 (2.61)	0.6259 (2.23)	-0.1082 (-2.74)
Redeployability × foreclosure cost	0.8076 (3.22)	0.0146 (0.30)	0.0080 (0.42)	-1.9448 (-1.75)	-0.1099 (-1.68)	0.6226 (2.88)
PANEL B: INTERACTIONS WITH STRICTNESS OF ZONING						
Redeployability	1.0669 (1.94)	-0.6668 (-2.66)	-0.4448 (-4.82)	-7.5846 (-1.14)	-2.6489 (-2.85)	0.3330 (2.01)
Redeployability × zoning strictness	-2.8903 (-2.39)	0.3669 (3.08)	0.2270 (5.39)	4.7521 (1.59)	1.6350 (3.75)	-0.1862 (-2.39)
Redeployability	1.1971 (0.57)	-0.2924 (-0.65)	-0.1370 (-0.82)	-15.4856 (-1.47)	-3.3267 (-2.13)	-0.2724 (-1.03)
Redeployability × growth management	-0.4061 (-1.89)	0.0797 (1.81)	0.0369 (2.02)	4.0564 (1.74)	0.9022 (2.60)	0.0512 (0.87)
PANEL C: INTERACTIONS WITH LOCAL MARKET LIQUIDITY						
Redeployability	0.2670 (0.91)	-0.3969 (-2.49)	-0.3000 (-4.74)	-8.5374 (-1.95)	-1.8720 (-3.09)	0.1501 (1.37)
Redeployability × demand-to-supply	-1.2878 (-1.64)	0.2108 (3.34)	0.1364 (5.79)	4.4819 (2.79)	1.1029 (4.73)	-0.0843 (-2.01)

Regression results of the loan interest rate, frequency of debt, total leverage, debt maturity, loan duration, and frequency of multiple creditors on asset redeployability (measured by the intensity of allowable use from the property's zoning designation) and its interaction with characteristics of the local market in which the property resides are reported. Panel A considers the interaction with ease of foreclosure at the state level. This variable is the sum of a judicial-foreclosure-only dummy and an index for the 1998 Fannie Mae optimum foreclosure time frame. Panel B examines interactions with variables designed to capture the strictness of zoning. The first is an index of zoning strictness which is the average of the following two measures from the Wharton Land Use Control Survey: the percentage of applications for zoning changes that were approved in the local MSA during 1989, and the estimated number of months between application for rezoning and issuance of a building permit for the development of a property in the MSA (average for single family units and office buildings). The second measure is an index of the effectiveness of growth management techniques employed in the MSA obtained from the Wharton Land Use Control Survey. Specifically, survey respondents' assessment of the effectiveness of ordinances, building permits, and zoning ordinances in controlling growth are provided on a scale of 1 (not important) to 5 (very important) and the average across the three categories is the growth management index. Panel C examines the interaction of redeployability with a measure of local market liquidity, namely, the average of the quantitative ratings of survey respondents from the Wharton Land Use Control Survey on the ratio of demand for land uses relative to the acreage of land zoned for those uses across single family, multifamily, commercial, and industrial uses, and across various lot sizes. All regressions include the regressors from Table II Panel A, including census tract, general zoning category, property type, and year fixed effects with robust standard errors. Appendix 2 details the sources and computations of all relevant variables.

Table IV

HISTORIC ZONING DESIGNATIONS						
Dependent variable =	Interest rate	Debt frequency	Leverage	Debt maturity	Loan duration	Multiple creditors
Historic	0.5913 (3.17)	-0.1085 (-2.27)	-0.0490 (-2.17)	-0.4942 (-0.38)	-0.1109 (-1.95)	0.1187 (2.49)
Redeployability	0.8540 (1.88)	-0.1205 (-1.31)	-0.0996 (-0.80)	-3.6482 (-0.83)	0.6445 (1.69)	0.2027 (1.56)
Redeployability × historic	-1.9869 (-3.84)	0.2219 (2.03)	0.3050 (2.26)	11.2079 (2.17)	0.3075 (0.59)	-0.3126 (-2.02)

Regression results of the loan interest rate, frequency of debt, total leverage, debt maturity, loan duration, and frequency of multiple creditors on an indicator variable for properties with a historic zoning designation are reported. Results from interacting the redeployability measure with the historic dummy are also reported. The regressions include the control variables from Table II, Panel A, including fixed effects for census tract, general zoning category, property type, and year with robust standard errors. Coefficient estimates and their associated *t*-statistics (in parentheses) are reported.

(though it is insignificant for duration) when relative demand is strong.

6 Short summary

- **Method.** Use property-specific zoning flexibility as an ex ante proxy for liquidation value in commercial real estate and regress loan terms on this proxy using very fine local fixed effects.
- **Identification.** Compare properties within the same census tract and broad zoning category, while controlling for current sale price, profitability, property type, year, and eventually bank identity.
- **Main result.** More redeployable assets receive lower-rate, larger, longer loans and are less likely to be financed by multiple creditors.

- **Validation.** The effect is weaker when foreclosure is difficult and stronger when zoning rules bind more tightly and when local land markets are tighter.
- **Bottom line.** Collateral quality — measured by liquidation value in the lender’s hands — materially affects financial contracting.

7 Conclusion

7.1 Core conclusion

The paper provides unusually direct evidence that **liquidation value affects debt contracts**. In a setting with asset-level financing and a plausible ex ante proxy for collateral redeployability, the data line up closely with incomplete-contracting theory.

7.2 Economic intuition

A lender cares about two things:

1. the asset’s ability to generate cash flow in the borrower’s hands;
2. the asset’s value if the lender must seize and liquidate it.

This paper isolates the second object. More flexible zoning raises the collateral’s value in bad states, so the lender is willing to lend more, for longer, at a lower promised yield, and with less need for multiple-creditor discipline.

7.3 Takeaway

This paper is important because it shifts attention from **how much collateral there is** to **how good that collateral is**. The empirical lesson is simple: when assets are easier to redeploy, they support more favorable financial contracts. That is exactly what incomplete-contracting and transaction-cost theories predict.